

Phase 3: Data Cleaning and Using Oracle

1. Dataset Description

Dataset 1 - IMDb: <https://github.com/sahildit>

The IMDb dataset contains 300,000+ movies ranging from release years 1894 to 2020, including titles, languages, genres, and production casts for each. The specific attributes that we plan to use from this dataset are the title, genre, year, votes, avg_vote, and imdb_title_id. This dataset will be used to answer all 3 questions by providing the names and information of the movies, allowing for the integration of the other datasets. Specifically for question 2, we will be using the avg_votes attribute in this dataset and comparing it to variables from other datasets to answer the question.

Dataset 2 - Reddit: https://huggingface.co/datasets/ZhankuiHe/reddit_movie_small_v1

This dataset contains Reddit discussions during the full year of 2022 about movie recommendations from five subreddits (r/movies, r/moviesuggestions, r/bestofnetflix, r/netflixbestof, r/truefilm) and has been processed to identify movie mentions and link them to IMDb IDs. This dataset contains 420,000 rows, where each row represents one conversational turn within a recommendation dialogue. This dataset will support all three research questions. After cleaning, it provides the number of recommendations per movie for Questions 1 and 2, and the number of upvotes analyzed in Question 3.

Dataset 3 - Box Office Mojo:

https://github.com/ntdoris/movie-revenue-analysis/blob/main/zippedData/bom.movie_gross.csv.gz

This dataset from Box Office Mojo contains data from approximately 4000 movies, with release dates ranging from 2010 to 2018. Key attributes in this dataset consist of domestic_gross, which refers to the combined Canada and USA box office gross, and foreign_gross, which refers to the combined box office gross from the rest of the world. This dataset will provide information on the domestic and foreign gross, which will be used to answer the first two research questions.

2. Describe the data cleaning steps you performed in Step 1 from the Workflow section. If you did not perform any data cleaning, briefly explain why.

IMDb dataset:

We only retained the variables relevant to our analysis: imdb_title_id, title, year, genre, duration, avg_vote, votes, and language. The dataset was filtered to only include English-language films released between 2010 and 2018. These years were chosen as two of the research questions' queries require the joining of this dataset and the Box Office Mojo dataset, and the Box Office Mojo dataset only contains movies from this timeline. Only movies whose genres include horror, action, or comedy were included, as these are the 3 genres the research questions address. We

then removed rows with missing values in the key variables. We then standardized movie titles by converting them to lowercase, trimming extra spaces, and removing unnecessary punctuation. This was done because this dataset would be joined with the Box Office Mojo dataset using the movie titles, thus both datasets needed to contain consistent, standardized names. Lastly, the year variable was converted to an integer format, and duplicate entries were removed to ensure that each `imdb_title_id` only appears once in the cleaned dataset.

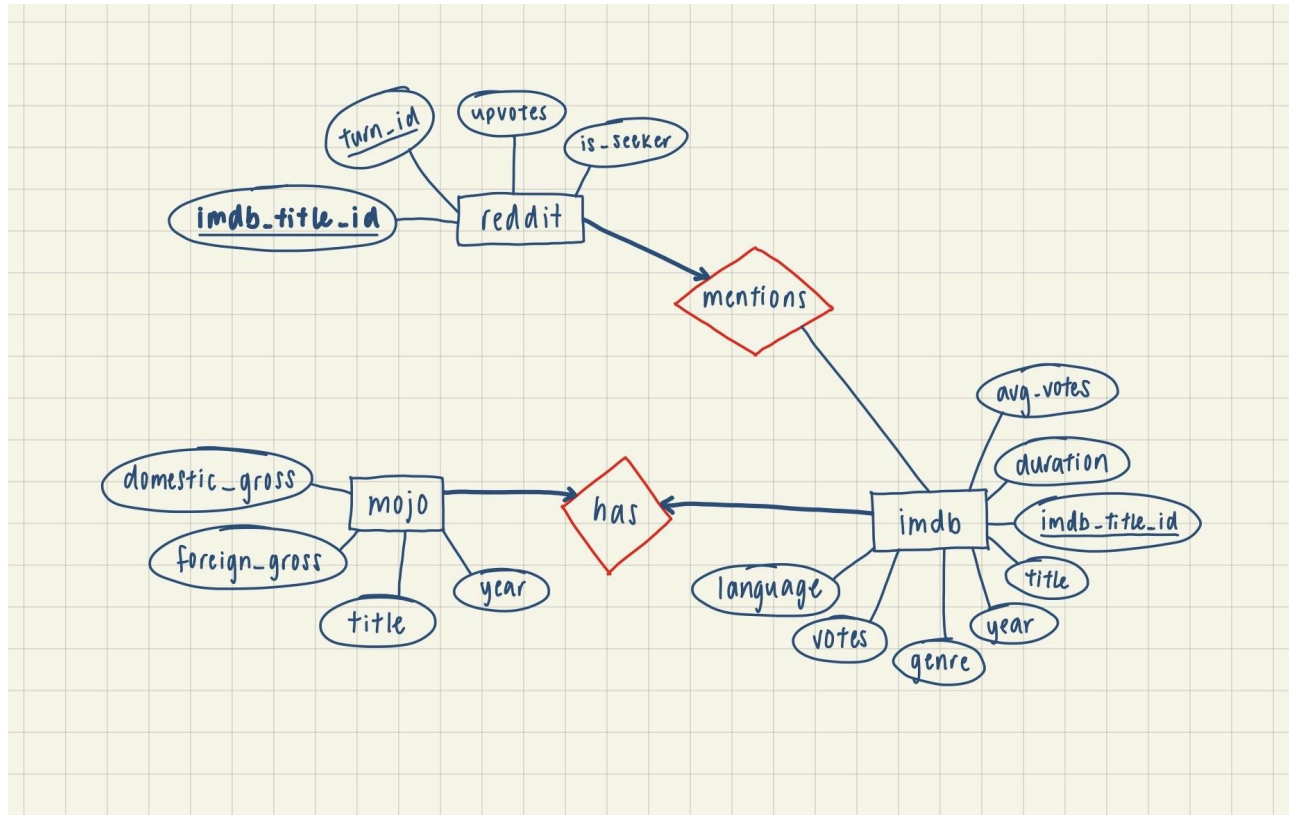
Reddit dataset:

We started cleaning the Reddit data by keeping only the attributes relevant to this project's research questions and analysis. The attributes kept were `turn_id`, `upvotes`, `is_seeker`, and the processed text that contains IMDb IDs, saved as `imdb_title_id`. Using this data, any row containing a null value was dropped as it was deemed unusable to the analysis. The original `processed_id` column contained IMDb IDs, which were needed in further SQL analysis, as it let this dataset be joined onto the IMDb dataset. For this reason, `processed_id` was scanned for its IMDb ID(s) (patterns like "tt1234567") and created a new table where each row represents one movie mention containing the attributes listed above. Lastly, we removed all duplicate rows containing the same combination of `turn_id` and `imdb_title_id`, as this would represent a movie being mentioned multiple times in one posting rather than a new recommendation.

Box Office Mojo dataset:

For the BOM dataset, we kept the columns movie title, year, domestic gross, and foreign gross. We then renamed the movie column to title, and standardized titles the same way as in IMDb (lowercased, trimmed spaces, removed extra punctuation) for easier joining with the IMDb dataset later on. We converted the year column to a numeric data type, filtered to movies released between 2010 and 2018 to match the year range in IMDb. The year, domestic gross, and foreign gross columns were converted to numeric values. After that, we dropped rows that were missing the title or year, and removed rows where both gross values were missing, as those rows could not be used to answer the questions that required box office values. We also filtered out rows where both gross values were negative. Finally, we sorted by title and year and removed duplicates so that there is at most one BOM record per (title, year) pair in the cleaned file.

3. ER diagram from Step 2 of the Workflow section.



4. State your overarching research question for your project. If you wish, you can reuse what you had for prior phases without rewording it.

Overarching research question: How does movie discussion relate to commercial success?

5. For each research question:

- State the research question.
- Provide the SQL query that answers it.
- Provide the file name of the Python file that contains the code that executes the query and include the exact line number where we can find the query in the code.
- Briefly explain what the query does and why it's appropriate.
- Present the result that is produced by your code (this could be a table and/or a visualization).

RQ1: How does Reddit recommendation frequency relate to the proportion of a movie's box office revenue earned internationally versus domestically among the movie genres horror, action, and comedy?

Python file: phase3_analysis.py, line 153

```

SELECT
    im.imdb_title_id,
    im.title,
    im.year,
    im.genre,
    NVL(r.num_recommendations, 0) AS num_recommendations,
    b.domestic_gross,
    b.foreign_gross,
    (b.domestic_gross + b.foreign_gross) AS total_revenue,
    (b.foreign_gross / NULLIF(b.domestic_gross + b.foreign_gross, 0)) AS foreign_share
FROM imdb_movies im
JOIN bom_gross b
    ON im.title = b.title
    AND im.year = b.year
LEFT JOIN (
    SELECT
        Imdb_title_id,
        COUNT(*) AS num_recommendations
    FROM reddit_mentions
    WHERE imdb_title_id IS NOT NULL
    GROUP BY imdb_title_id) r
    ON im.imdb_title_id = r.imdb_title_id
WHERE
    im.genre LIKE '%Horror%' OR
    im.genre LIKE '%Action%' OR
    im.genre LIKE '%Comedy%'
ORDER BY num_recommendations DESC;

```

Here is the output from SQL:

```

tt2126355
san andreas
2015

IMDB_TITLE_ID
-----
TITLE
-----
YEAR
-----
GENRE
-----
NUM_RECOMMENDATIONS DOMESTIC_GROSS FOREIGN_GROSS TOTAL_REVENUE FOREIGN_SHARE
-----
Action, Adventure, Drama
27 155190832 301067707 456258539 .659862077

IMDB_TITLE_ID
-----
TITLE
-----
YEAR
-----
GENRE
-----
NUM_RECOMMENDATIONS DOMESTIC_GROSS FOREIGN_GROSS TOTAL_REVENUE FOREIGN_SHARE
-----
tt2788710
the interview
2014

IMDB_TITLE_ID
-----
TITLE
-----
YEAR
-----
GENRE
-----
NUM_RECOMMENDATIONS DOMESTIC_GROSS FOREIGN_GROSS TOTAL_REVENUE FOREIGN_SHARE
-----
Action, Adventure, Comedy
27 6105175 6237457 12342632 .505358744

```

Total output: 448 rows selected.

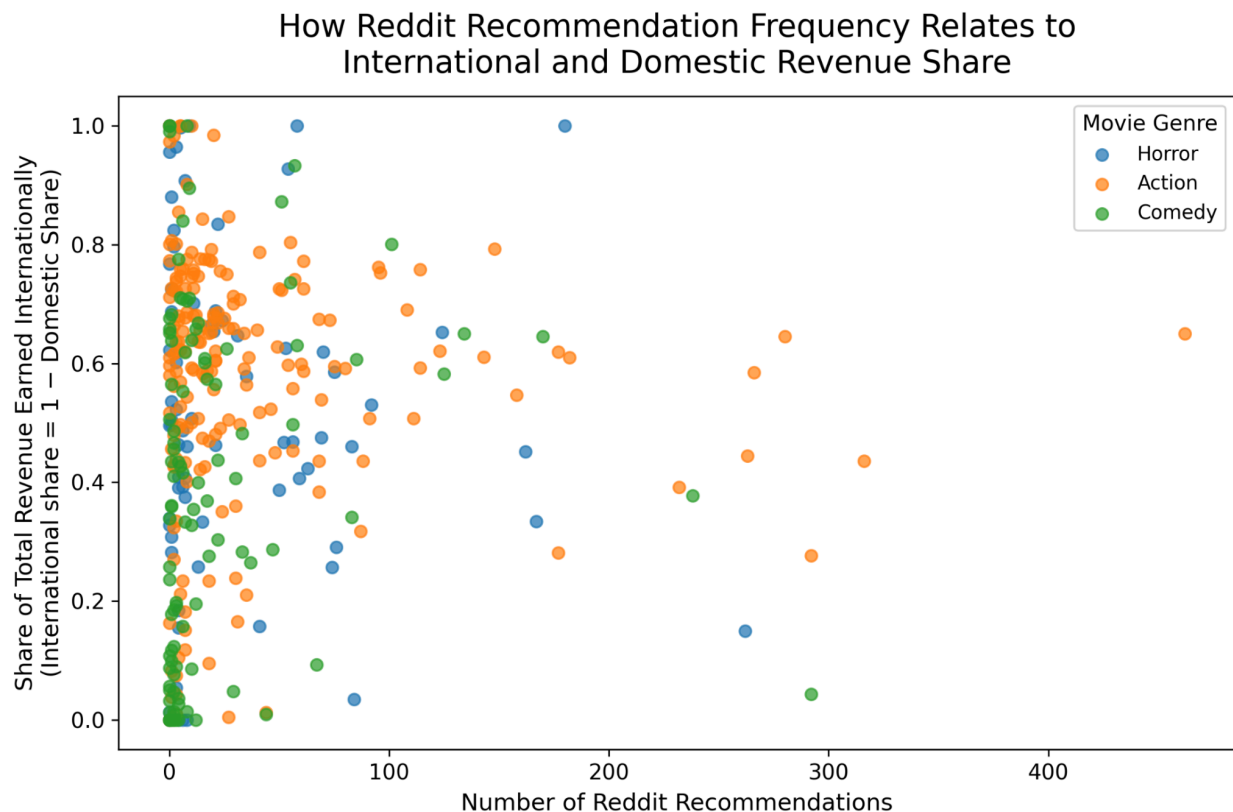
SOL query description

This query combines information from the three datasets to analyze the relationship between Reddit recommendations and international revenue share for movies in the horror, action, and comedy genres. It joins the imdb_movies table with the bom_gross table using the title and year to match movies with their box office revenue. A subquery aggregates the reddit_mentions table by the imdb_title_id to calculate the number of Reddit recommendations for each movie. The results are then merged with the movie data using a left join, ensuring that movies without Reddit mentions are still included in the data set. The NVL() function replaces missing recommendations with a count of 0.

The query calculates the total revenue by adding the domestic_gross value with the foreign_gross value, while the foreign share is calculated by dividing foreign_gross and

total_revenue. The NULLIF() function ensures there are no division-by-zero errors when calculating foreign_share. Finally, the query is filtered to our genres of interest.

This query is appropriate because it returns the key variables needed to analyse the research question: Reddit recommendation counts, foreign revenue share, and genre. Performing the joins, aggregations, and calculations in SQL ensures that datasets can be directly used in Python for visualisation, Pearson correlation analysis, and multiple linear regression.



The scatterplot above shows the relationships between Reddit recommendation frequency and the share of total revenue earned internationally across horror, action, and comedy films. Most movies have less than 50 recommendations, indicating that highly discussed films are relatively rare. There is no strong linear relationship between recommendation frequency and foreign revenue share, as the points are widely dispersed vertically. However, action films tend to appear more frequently among higher recommendation counts and show high international revenue shares. Across the genres of interest, foreign revenue share commonly falls between 0.4 and 0.8, regardless of Reddit activity. Since domestic revenue share is the complement of foreign revenue share (i.e. domestic revenue share = $1 - \text{foreign revenue share}$), this suggests that Reddit recommendation frequency is not strongly associated with both a movie's share of domestic revenue and its share of international revenue.

RQ2: How does the average IMDb vote on a movie relate to the number of Reddit recommendations versus the box office success?

Python file: phase3_analysis.py, line 297

```
SELECT i.avg_vote, (b.domestic_gross + b.foreign_gross) AS total_gross, COUNT(*) AS
reddit_discussion_count
FROM imdb_movies i
JOIN reddit_mentions r
      ON i.imdb_title_id = r.imdb_title_id
JOIN bom_gross b
      ON LOWER(i.title) = LOWER(b.title)
      AND i.year = b.year
GROUP BY i.title, i.imdb_title_id, b.domestic_gross, b.foreign_gross, i.avg_vote;
```

Here is the output from SQL:

AVG_VOTE	TOTAL_GROSS	REDDIT_DISCUSSION_COUNT
6.3	38771262	21
7.1	335802233	5
5.7	82182803	1
5.6	1703281	5
5.9	142531552	8
6.6	561072059	29
4.1	35688	2
5.4	2791973	3
7.6	84393749	526
7.4	59596747	232
7.5	16553285	292
AVG_VOTE	TOTAL_GROSS	REDDIT_DISCUSSION_COUNT
6.1	12758799	15

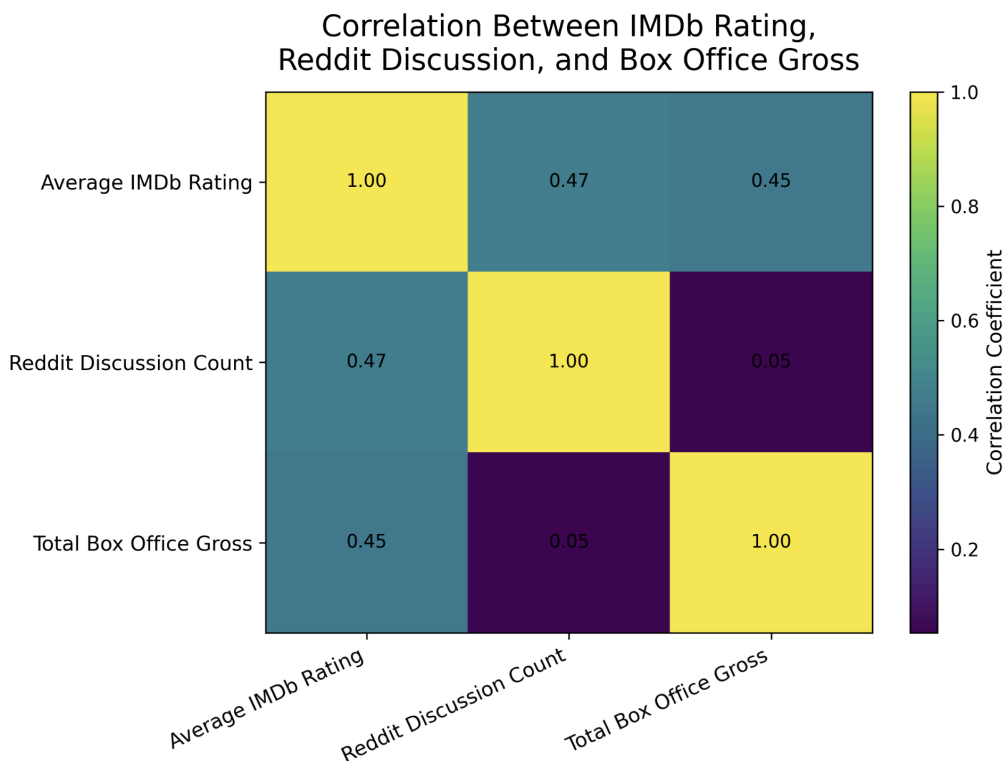
342 rows selected.

SQL Query Description

This query combines the IMDb, Reddit, and Box Office Mojo datasets to create a movie-level dataset containing each film's average IMDb rating, total box office gross, and Reddit discussion count. The `imdb_movies` and `reddit_mentions` tables are joined by `imdb_title_id`, while `imdb_movies` is joined with `bom_gross` using a case-insensitive match on title and exact match on year. The `total_gross` attribute is calculated by summing the domestic gross and the foreign

gross per movie. The query makes another column by grouping by movie title, IMDb title ID, domestic gross, foreign gross, and average vote, and then calculates the total count of Reddit posts per movie. This count is saved as the reddit_discussion_count. The result of this query returns the average IMDb vote, total gross, and Reddit discussion count for each movie.

This query is appropriate for this research question as the data will be used to run linear regressions to calculate both total_gross vs the IMDb avg_vote and total_gross vs reddit_discussion_count. A multiple linear regression of total_gross vs reddit_discussion_count and IMDb avg_vote will also be run to investigate how the two variables jointly relate to the total_gross. As well, the resulting dataset can be used in Python to compute values such as R^2 and generate a correlation matrix. Comparing these results, including the R^2 values, helps evaluate whether IMDb average vote, Reddit discussion count, or both together are more strongly correlated with box office success.



The heatmap above shows the correlations among the three variables in the resulting SQL query, providing insight into their relationships. From this visualization, it can be seen that the average IMDb rating and the total box office success correlate at 0.45, which is much higher than the 0.05 correlation between the count of Reddit discussions and total box office success. These correlations indicate that the average IMDb rating may be significant in influencing a movie's total box office success, however further analysis will be done on each of the relationships to confirm this.

RQ3: How does the average number of upvotes for Reddit recommendations and discussion posts vary by movie duration, and how does this relationship change based on movie genre (horror, action or comedy)?

Python file: phase3_analysis.py, line 408

```
SELECT duration_bins, genre_filtered, AVG (upvotes) as avg_upvotes
FROM
    (SELECT r.upvotes,
        CASE
            WHEN i.duration >= 60 AND i.duration <80 THEN '60-79'
            WHEN i.duration >= 80 AND i.duration <100 THEN '80-99'
            WHEN i.duration >= 100 AND i.duration <120 THEN '100-119'
            WHEN i.duration >= 120 AND i.duration <140 THEN '120-139'
            WHEN i.duration >= 140 AND i.duration <160 THEN '140-159'
            WHEN i.duration >= 160 AND i.duration <180 THEN '160-179'
            WHEN i.duration >= 180 AND i.duration <=200 THEN '180-200'
            ELSE 'N/A'
        END AS duration_bins,
        CASE
            WHEN i.genre LIKE '%Horror%' THEN 'Horror'
            WHEN i.genre LIKE '%Action%' THEN 'Action'
            WHEN i.genre LIKE '%Comedy%' THEN 'Comedy'
        END AS genre_filtered
    FROM imdb_movies i
    JOIN reddit_mentions r ON i.imdb_title_id = r.imdb_title_id
    WHERE
        i.genre LIKE '%Horror%' OR
        i.genre LIKE '%Action%' OR
        i.genre LIKE '%Comedy%')
GROUP BY duration_bins, genre_filtered
HAVING AVG(upvotes) IS NOT NULL;
```

SQL Query Description

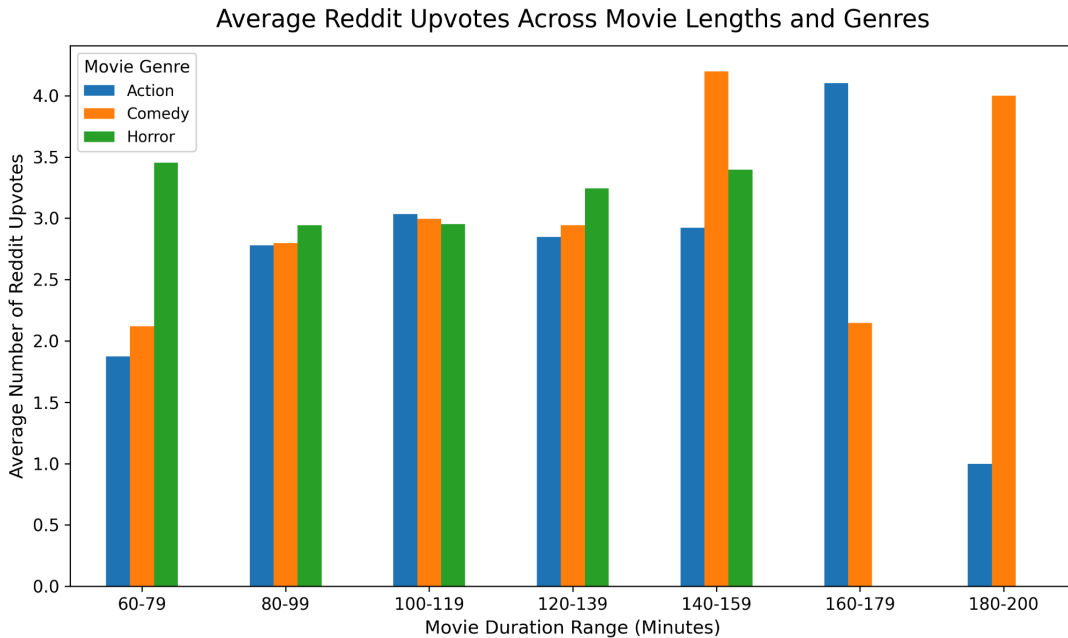
This query answers the question by grouping movie durations, which range from 60-200 minutes, into bins in increments of 20 minutes, then filtering for the three genres of interest, and calculating the average number of upvotes for posts in each genre category. The query uses the IMDb and Reddit datasets, joined by using the IMDb title IDs, resulting in a dataframe where each row corresponds with a Reddit post. The query cleans the movie genre attribute by replacing any lists of genres with the first relevant genre (either horror, action, or comedy), then

filters only for movies that are of the three select genre types. Next, it groups by the binned durations and genre, then calculates and returns the average number of upvotes across all Reddit posts within that subset that are not null values.

There are a couple of caveats to this query approach. The genre attribute in the IMDb dataset is coded such that movies that cover multiple genres have all three listed, and separated by commas. As such, in order to prevent duplication of the same movie title, the query we created categorizes movies once by the first genre listed (if there are multiple genres).

Here is the output from SQL:

DURATIO	GENRE_	AVG_UPVOTES
-----	-----	-----
140-159	Action	2.92267267
160-179	Action	4.1004902
80-99	Horror	2.94424406
100-119	Action	3.03298376
100-119	Comedy	2.99564033
100-119	Horror	2.95446698
120-139	Horror	3.24313023
140-159	Horror	3.39779006
60-79	Horror	3.45274725
60-79	Comedy	2.12021858
120-139	Comedy	2.94551845
DURATIO	GENRE_	AVG_UPVOTES
-----	-----	-----
80-99	Action	2.78049713
120-139	Action	2.84772647
160-179	Comedy	2.14814815
180-200	Action	1
180-200	Comedy	4
80-99	Comedy	2.79732382
140-159	Comedy	4.1971831
60-79	Action	1.875



The bar graph above visualizes the output of the SQL query, including the average number of Reddit upvotes across different movie lengths and the genres of action, comedy, and horror. The absence of green bars for the horror genre in the movie duration bins of 160-179 minutes and 180-200 minutes is likely because there are no Reddit posts with one or more upvotes for horror movies within those ranges. The bar graph seems to indicate that there are specific genre and movie duration combinations that result in a higher number of average upvotes. For example, comedy movies that have a duration within 140-159 minutes have a noticeably higher number of average upvotes compared to other genres in the same duration category.

6. AI Tool Use Declaration: Refer to the Project Overview document for expectations.

AI identified an issue with the quotations used in the SQL queries. It informed us that the quotations that were originally used did not work and needed to be re-written in SQLPlus. It suggested using the NVL() function to replace null values in the recommendation count with 0. We also asked it to identify which SQL function we should use to prevent issues with zero division when calculating foreign shares.

AI Tool: ChatGPT

Prompt: I am writing a SQL query using Oracle and I want to turn null values into 0. What SQL function should I use?

Response:

In **Oracle SQL**, the classic function for turning **NULL** into something else (like **0**) is **NVL()**.

Syntax

</> SQL

```
NVL(expression, replacement_value)
```

If **expression** is **NULL**, Oracle returns **replacement_value**.

Example

</> SQL

```
SELECT NVL(sales, 0) AS sales
FROM orders;
```

If **sales** is **NULL**, this will return **0**.

AI Tool: Claude

Transcript

Attached to these messages was the photo of the SQL query error:

```
SQL> SELECT
  2   im.imdb_title_id,
  3   im.title,
  4   im.year,
  5   im.genre,
  6   NVL(r.num_recommendations, 0) AS num_recommendations,
  7   b.domestic_gross,
  8   b.foreign_gross,
  9   (b.domestic_gross + b.foreign_gross) AS total_revenue,
 10   (b.foreign_gross / (b.domestic_gross + b.foreign_gross)) AS foreign_share
 11 FROM imdb_movies im
 12 JOIN bom_gross b
 13   ON im.title = b.title
 14   AND im.year = b.year
 15 LEFT JOIN (
 16   SELECT
 17     Imdb_title_id,
 18     COUNT(*) AS num_recommendations
 19   FROM reddit_mentions
 20   WHERE imdb_title_id IS NOT NULL
 21   GROUP BY imdb_title_id) r
 22   ON im.imdb_title_id = r.imdb_title_id
 23 WHERE
 24   im.genre LIKE '%Horror%' OR
 25   im.genre LIKE '%Action%' OR
 26   im.genre LIKE '%Comedy%'
 27 ORDER BY num_recommendations DESC;
        im.genre LIKE '%Horror%' OR
        *
```

ERROR at line 24:
ORA-00911: invalid character
Help: <https://docs.oracle.com/error-help/db/ora-00911/>

why is this?

I can see `LIKE` is still there on line 25! Change it to `LIKE`:

```
sql
im.genre LIKE '%Action%' OR
```



but it says line 24

The error points to line 24 but it's actually the quotes — those are **smart quotes** ' instead of regular straight quotes '. This happens when you type in Word or some text editors.

Make sure all your quotes look like ' not ' or '. Retype the quotes manually in SQLPlus or a plain text editor like Notepad.

AI Tool: Claude

Prompt: Why is this `b.foreign_gross / (b.domestic_gross + b.foreign_gross) AS foreign_share` not working?

Answer:

Some movies have `domestic_gross + foreign_gross = 0`, so it's dividing by zero. Wrap it in a `NULLIF` to handle that:

```
sql
(b.foreign_gross / NULLIF(b.domestic_gross + b.foreign_gross, 0)) AS foreign_share
```

That'll return `NULL` instead of crashing when the total is zero.